Experiment 20 — RentRide: Online Vehicle Rental System

Aim:

To develop a responsive online vehicle rental website that allows users to search vehicles, book rentals, view booking status, manage profiles, and submit reviews, while administrators manage vehicles and bookings.

Algorithm

1. Start the RentRide website and load vehicle, profile, booking, and review data from browser storage.
2. Allow users to search vehicles by name, type, location, and daily rental price.
3. Let the user select a vehicle, enter customer details, choose pickup and return dates, and calculate the rental cost.
4. Validate the booking information and check for overlapping booking requests.
5. Save valid booking requests and display their status in the user's booking dashboard.
6. Allow the administrator to add or delete vehicles, approve or reject bookings, update booking status, and export booking records.
7. Save the updated information in browser storage and refresh the displayed data.

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>RentRide | Vehicle Rentals</title>

<style>

:root{--navy:#10233f;--blue:#2563eb;--bg:#f3f6fb;--green:#16a34a}
```

```
*{box-sizing:border-box}
body{margin:0;font-family:Arial,sans-serif;background:var(--bg);color:#18243a}
header{background:var(--navy);color:white;padding:18px 6%;display:flex;
justify-content:space-between;align-items:center;flex-wrap:wrap;gap:12px}
.logo{font-size:25px;font-weight:bold;color:#93c5fd}
nav{display:flex;gap:8px;flex-wrap:wrap}
button{border:0;border-radius:9px;padding:10px 15px;cursor:pointer;
font-weight:bold;background:var(--blue);color:white}
button:hover{opacity:.85}
nav button{background:#263d5d}
button.secondary{background:#e2e8f0;color:#18243a}
button.danger{background:#dc2626}
button.success{background:var(--green)}
.hero{padding:38px 6%;background:linear-gradient(120deg,#10233f,#2563eb);
color:white}
.hero h1{font-size:clamp(28px,5vw,46px);margin:0 0 12px}
.hero p{color:#dbeafe}
.container{max-width:1200px;margin:25px auto;padding:0 18px}
.panel{background:white;padding:20px;border-radius:15px;margin-bottom:22px;
box-shadow:0 5px 20px #15294b0b}
.filters,.formgrid{display:grid;grid-template-columns:repeat(auto-
fit,minmax(150px,1fr));gap:12px}
input,select,textarea{width:100%;padding:11px;border:1px solid #d5deeb;
border-radius:8px;font:inherit;background:white}
label{font-size:13px;font-weight:bold;display:block;margin-bottom:6px}
.field{margin-bottom:12px}
h2{margin-top:0}
.cards{display:grid;grid-template-columns:repeat(auto-
fit,minmax(255px,1fr));gap:18px}
```

.card{background:white;border-radius:15px;overflow:hidden;box-shadow:0 5px 20px #15294b10}

.carimg{height:165px;background:#dbeafe;display:flex;align-items:center;

justify-content:center;font-size:75px}

.cardbody{padding:17px}

.tag{display:inline-block;background:#dbeafe;color:#1d4ed8;padding:5px 9px;

border-radius:20px;font-size:12px;margin:3px 0}

.price{font-size:24px;font-weight:bold;color:var(--blue)}

.muted{color:#64748b;font-size:13px}

.row{display:flex;align-items:center;justify-content:space-between;gap:10px;flex-wrap:wrap}

table{width:100%;border-collapse:collapse;min-width:650px}

th,td{text-align:left;padding:12px;border-bottom:1px solid #e2e8f0;font-size:14px}

th{background:#eff6ff}

.tablewrap{overflow-x:auto}

.status{padding:5px 8px;border-radius:7px;background:#fef3c7;color:#92400e;font-size:12px}

.hidden{display:none!important}

.statgrid{display:grid;grid-template-columns:repeat(auto-fit,minmax(130px,1fr));gap:12px}

.stat{background:#eff6ff;border-radius:12px;padding:16px}

.stat strong{font-size:25px;display:block;margin-top:8px}

footer{text-align:center;padding:24px;background:var(--navy);color:white;margin-top:35px}

#toast{position:fixed;bottom:20px;right:20px;background:#10233f;color:white;

padding:14px 18px;border-radius:10px;display:none;z-index:10;max-width:90%}

@media(max-width:600px){header{padding:15px}.hero{padding:28px 5%}.container{padding:0 12px}}

</style>

</head>

```
<body>
<header>
 <div class="logo">🚗 RentRide</div>
 <nav>
  <button onclick="showPage('home')">Home</button>
  <button onclick="showPage('bookings')">My Bookings</button>
  <button onclick="showPage('profile')">My Profile</button>
  <button onclick="showPage('reviews')">Reviews</button>
  <button onclick="showPage('admin')">Admin</button>
 </nav>
</header>

<section class="hero">
 <h1>Your journey. Your vehicle.</h1>
 <p>Find your perfect ride and book it in minutes.</p>
 <button
onclick="document.getElementById('vehicles').scrollIntoView({behavior:'smooth'})">
  Explore Vehicles ↓
 </button>
</section>

<main class="container">
<section id="home">
 <div class="panel">
  <h2>Find your ride</h2>
  <div class="filters">
   <div><label>Search vehicle</label><input id="search" placeholder="Car, SUV,
truck..." oninput="renderCars()"></div>
```

```
    <div><label>Vehicle type</label>
     <select id="type" onchange="renderCars()">
      <option value="">All types</option><option>Car</option><option>SUV</option>
      <option>Truck</option><option>Bike</option>
     </select>
    </div>
    <div><label>Location</label><input id="location" placeholder="e.g. Chennai" oninput="renderCars()"></div>
    <div><label>Maximum price/day (₹)</label><input id="maxprice" type="number" min="0" placeholder="Any price" oninput="renderCars()"></div>
   </div>
  </div>
  <div class="row"><h2 id="vehicles">Available vehicles</h2><span class="muted" id="carcount"></span></div>
  <div class="cards" id="carlist"></div>
</section>

<section id="bookings" class="hidden">
  <div class="panel">
   <div class="row"><h2>My bookings</h2><button class="secondary" onclick="showPage('home')">+ Book a vehicle</button></div>
   <div class="tablewrap"><table><thead><tr><th>Booking ID</th><th>Vehicle</th><th>Dates</th><th>Total</th><th>Status</th><th>Action</th></tr></thead>
    <tbody id="mybookingrows"></tbody></table></div>
  </div>
</section>

<section id="profile" class="hidden">
  <div class="panel">
```

<h2>Customer profile</h2>

Enter your details to attach them to your bookings.

<label>Full name</label><input id="pname" placeholder="Your name">

<label>Phone number</label><input id="phone" placeholder="10-digit number">

<label>Email</label><input id="email" type="email" placeholder="you@example.com">

<label>Driving licence number</label><input id="licence" placeholder="Licence number">

<button onclick="saveProfile()">Save profile</button>

Demo application: profile data is stored in this browser only.

</section>

<section id="reviews" class="hidden">

<h2>Customer reviews</h2>

<label>Your name</label><input id="rname" placeholder="Name">

<label>Rating</label><select id="rating"><option value="5">★★★★★ (5)</option><option value="4">★★★★ (4)</option><option value="3">★★★ (3)</option><option value="2">★★ (2)</option><option value="1">★ (1)</option></select>

<label>Review</label><textarea id="reviewtext" rows="3" placeholder="Share your rental experience"></textarea>

<button onclick="addReview()">Submit review</button>

```
 </div>
 <div id="reviewlist" class="cards"></div>
</section>

<section id="admin" class="hidden">
 <div class="panel">
  <h2>Administrator dashboard</h2>
  <p class="muted">Demo admin password: <b>admin123</b>. This is not secure authentication.</p>
  <div id="adminlogin">
   <div class="field"><label>Admin password</label><input id="adminpass" type="password" placeholder="Enter demo password"></div>
   <button onclick="adminSignIn()">Login as admin</button>
  </div>
  <div id="adminarea" class="hidden">
   <div class="statgrid">
    <div class="stat">Vehicles<strong id="statcars">0</strong></div>
    <div class="stat">Bookings<strong id="statbookings">0</strong></div>
    <div class="stat">Pending<strong id="statpending">0</strong></div>
   </div>
   <hr style="border:0;border-top:1px solid #e2e8f0;margin:22px 0">
   <h3>Add a vehicle</h3>
   <div class="formgrid">
    <div class="field"><label>Vehicle name</label><input id="aname" placeholder="e.g. City Cruiser"></div>
    <div class="field"><label>Type</label><select id="atype"><option>Car</option><option>SUV</option><option>Truck</option><option>Bike</option></select></div>
    <div class="field"><label>Price per day (₹)</label><input id="aprice" type="number" min="1" placeholder="1500"></div>
```

<label>Location</label><input id="alocation" placeholder="Chennai">

<label>Seats</label><input id="aseats" type="number" min="1" value="5">

<label>Emoji/photo symbol</label><input id="aemoji" placeholder=" 🚙 " value=" 🚗 ">

<button onclick="addCar()">Add vehicle</button>

<h3 style="margin-top:28px">Manage vehicles</h3>

<table><thead><tr><th>Vehicle</th><th>Type</th><th>Price/day</th><th>Location</th><th>Action</th></tr></thead><tbody id="admincars"></tbody></table>

<h3 style="margin-top:28px">Manage bookings</h3>

<table><thead><tr><th>ID</th><th>Customer</th><th>Vehicle</th><th>Dates</th><th>Status</th><th>Update status</th></tr></thead><tbody id="adminbookings"></tbody></table>

<button class="secondary" onclick="exportBookings()">Export bookings CSV</button>

</section>

<section id="bookingform" class="hidden">

<h2>Book your vehicle</h2><button class="secondary" onclick="showPage('home')">✕ Close</button>

<label>Customer name</label><input id="bname" placeholder="Full name">

<label>Phone number</label><input id="bphone" placeholder="Phone number">

<label>Pickup date</label><input id="startdate" type="date">

<label>Return date</label><input id="enddate" type="date">

<label>Pickup location</label><input id="pickup" placeholder="Pickup address">

<label>Rental timing</label><select id="timing"><option>9 AM - 9 AM</option><option>10 AM - 10 AM</option><option>12 PM - 12 PM</option><option>6 PM - 6 PM</option></select>

Total: ₹0

Bookings are requests until approved by the administrator. Payment is not processed.

<button onclick="submitBooking()">Submit booking request</button>

</section>

</main>

<footer>© 2026 RentRide · Drive safely, travel freely.</footer>

<script>

const defaultCars=[

{id:1,name:"City Cruiser",type:"Car",price:1800,location:"Chennai",seats:5,emoji:"🚗",available:true},

{id:2,name:"Mountain SUV",type:"SUV",price:2800,location:"Bengaluru",seats:7,emoji:"🚙",available:true},

{id:3,name:"Urban Bike",type:"Bike",price:700,location:"Chennai",seats:2,emoji:"🏍️",available:true},

```
{id:4,name:"Cargo Truck",type:"Truck",price:3500,location:"Coimbatore",seats:3,emoji:"🚚",available:true},
{id:5,name:"Family Explorer",type:"SUV",price:2400,location:"Madurai",seats:7,emoji:"🚙",available:true},
{id:6,name:"Premium Sedan",type:"Car",price:2200,location:"Bengaluru",seats:5,emoji:"🚘",available:true}
];
function load(key,fallback){try{const v=localStorage.getItem(key);return v?JSON.parse(v):fallback}catch(e){return fallback}}
let cars=load("rr_cars",defaultCars);
let bookings=load("rr_bookings",[]);
let reviews=load("rr_reviews",[]);
let profile=load("rr_profile",{});
let chosen=null,adminLogged=false;

function save(){localStorage.setItem("rr_cars",JSON.stringify(cars));localStorage.setItem("rr_bookings",JSON.stringify(bookings));localStorage.setItem("rr_reviews",JSON.stringify(reviews));localStorage.setItem("rr_profile",JSON.stringify(profile))}
function toast(msg){const t=document.getElementById("toast");t.textContent=msg;t.style.display="block";setTimeout(()=>t.style.display="none",2600)}
function safe(s){return String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;","'":"&#39;"}[c]))}
function showPage(id){
 document.querySelectorAll("main>section").forEach(s=>s.classList.add("hidden"));
 document.getElementById(id).classList.remove("hidden");
 if(id==="bookings")renderBookings();
 if(id==="admin")renderAdmin();
```

```
 if(id==="reviews")renderReviews();
 window.scrollTo({top:0,behavior:"smooth"});
}
function renderCars(){
 const q=document.getElementById("search").value.toLowerCase();
 const ty=document.getElementById("type").value;
 const loc=document.getElementById("location").value.toLowerCase();
 const max=Number(document.getElementById("maxprice").value)||Infinity;
 const list=cars.filter(c=>c.available&&c.name.toLowerCase().includes(q)&&
 (!ty||c.type===ty)&&c.location.toLowerCase().includes(loc)&&c.price<=max);
 document.getElementById("carcount").textContent=list.length+" vehicles found";
 document.getElementById("carlist").innerHTML=list.map(c=>`
 <article class="card"><div class="carimg">${safe(c.emoji)}</div><div
class="cardbody">
 <div class="row"><h3>${safe(c.name)}</h3><span
class="tag">${safe(c.type)}</span></div>
 <p class="muted">📍 ${safe(c.location)} · 👥 ${c.seats} seats</p>
 <p class="price">₹${c.price.toLocaleString("en-IN")} <small class="muted">/
day</small></p>
 <button onclick="startBooking(${c.id})">Book this vehicle →</button>
 </div></article>`).join("")||'<div class="panel">No matching vehicles. Try changing your
filters.</div>';
}
function startBooking(id){
 chosen=cars.find(c=>c.id===id);if(!chosen)return;
 document.getElementById("selectedcar").textContent=chosen.emoji+"
"+chosen.name+" · ₹"+chosen.price+" per day";
 document.getElementById("bname").value=profile.name||"";
 document.getElementById("bphone").value=profile.phone||"";
```

```
 document.getElementById("pickup").value=chosen.location;
 const today=new Date();const date=today.toISOString().slice(0,10);
 document.getElementById("startdate").min=date;
 document.getElementById("enddate").min=date;
 document.getElementById("startdate").value=date;
 const tomorrow=new Date(today);tomorrow.setDate(tomorrow.getDate()+1);
 document.getElementById("enddate").value=tomorrow.toISOString().slice(0,10);
 showPage("bookingform");estimate();
}
function estimate(){
 if(!chosen)return;
 const s=new Date(document.getElementById("startdate").value+"T00:00:00");
 const e=new Date(document.getElementById("enddate").value+"T00:00:00");
 const days=Math.ceil((e-s)/86400000);
 document.getElementById("estimate").textContent="Total: ₹"+(days>0?days*chosen.price:0).toLocaleString("en-IN")+(days>0?" · "+days+" day(s)":" · Select valid dates");
}
document.getElementById("startdate").addEventListener("change",()=>{
 const s=document.getElementById("startdate").value;
 document.getElementById("enddate").min=s;
 if(document.getElementById("enddate").value<=s){
 const d=new Date(s+"T00:00:00");d.setDate(d.getDate()+1);
 document.getElementById("enddate").value=d.toISOString().slice(0,10);
 }estimate();
});
document.getElementById("enddate").addEventListener("change",estimate);

function submitBooking(){
```

```
 const name=document.getElementById("bname").value.trim();
 const phone=document.getElementById("bphone").value.trim();
 const start=document.getElementById("startdate").value;
 const end=document.getElementById("enddate").value;
 const pickup=document.getElementById("pickup").value.trim();
 if(!chosen||!name||!phone||!start||!end||!pickup)return toast("Please complete all booking fields.");
 if(!/^[0-9+()\-\s]{7,16}$/.test(phone))return toast("Enter a valid phone number.");
 if(end<=start)return toast("Return date must be after pickup date.");
 const days=Math.ceil((new Date(end+"T00:00:00")-new Date(start+"T00:00:00"))/86400000);
 const conflict=bookings.some(b=>b.carId===chosen.id&&["Approved","Pending"].includes(b.status)&&start<b.end&&end>b.start);
 if(conflict)return toast("This vehicle has a booking request for overlapping dates. Please choose other dates.");
 const booking={id:"RR"+Date.now().toString().slice(-8),carId:chosen.id,vehicle:chosen.name,
 name,phone,start,end,pickup,timing:document.getElementById("timing").value,
 days,total:days*chosen.price,status:"Pending",created:new Date().toLocaleString()};
 bookings.unshift(booking);profile={...profile,name,phone};save();
 toast("Booking request submitted! ID: "+booking.id);showPage("bookings");
}
function renderBookings(){
 const rows=bookings.filter(b=>!profile.name||b.name.toLowerCase()===profile.name.toLowerCase());
 document.getElementById("mybookingrows").innerHTML=rows.map(b=>`<tr>
 <td>${safe(b.id)}</td><td>${safe(b.vehicle)}</td><td>${safe(b.start)} → ${safe(b.end)}</td>
```

<td>₹${b.total.toLocaleString("en-IN")}</td><td><span class="status">${safe(b.status)}</td>

<td>${b.status==="Pending"?`<button class="danger" onclick="cancelBooking('${safe(b.id)}')">Cancel</button>`:"—"}</td></tr>`).join("")||'<tr><td colspan="6">No bookings yet. Book a vehicle to get started.</td></tr>';

}

function cancelBooking(id){

const b=bookings.find(x=>x.id===id);if(!b||b.status!=="Pending")return toast("Only pending bookings can be cancelled.");

b.status="Cancelled";save();renderBookings();toast("Booking cancelled.");

}

function saveProfile(){

profile={name:document.getElementById("pname").value.trim(),phone:document.getElementById("phone").value.trim(),email:document.getElementById("email").value.trim(),licence:document.getElementById("licence").value.trim()};

if(!profile.name)return toast("Please enter your name.");

save();toast("Profile saved.");renderBookings();

}

function addReview(){

const name=document.getElementById("rname").value.trim(),textval=document.getElementById("reviewtext").value.trim();

if(!name||!textval)return toast("Enter your name and review.");

reviews.unshift({name,rating:Number(document.getElementById("rating").value),text:textval,date:new Date().toLocaleDateString()});

save();document.getElementById("reviewtext").value="";renderReviews();toast("Review submitted.");

}

function renderReviews(){

```
document.getElementById("reviewlist").innerHTML=reviews.map(r=>`<article class="card"><div class="cardbody">

<div class="row"><b>${safe(r.name)}</b><span>⭐ ${r.rating}/5</span></div>

<p>${safe(r.text)}</p><span class="muted">${safe(r.date)}</span></div></article>`).join("")||'<div class="panel">Be the first to review RentRide.</div>';

}

function adminSignIn(){

if(document.getElementById("adminpass").value!=="admin123")return toast("Incorrect demo password.");

adminLogged=true;document.getElementById("adminlogin").classList.add("hidden");


document.getElementById("adminarea").classList.remove("hidden");renderAdmin();toast("Admin dashboard opened.");

}

function renderAdmin(){

if(!adminLogged)return;

document.getElementById("statcars").textContent=cars.length;

document.getElementById("statbookings").textContent=bookings.length;


document.getElementById("statpending").textContent=bookings.filter(b=>b.status==="Pending").length;


document.getElementById("admincars").innerHTML=cars.map(c=>`<tr><td>${safe(c.name)}</td><td>${safe(c.type)}</td>

<td>₹${c.price}</td><td>${safe(c.location)}</td><td><button class="danger" onclick="deleteCar(${c.id})">Delete</button></td></tr>`).join("");


document.getElementById("adminbookings").innerHTML=bookings.map(b=>`<tr><td>${safe(b.id)}</td>

<td>${safe(b.name)}</td><td>${safe(b.vehicle)}</td><td>${safe(b.start)} → ${safe(b.end)}</td><td>${safe(b.status)}</td><td>
```

```
 <select onchange="updateStatus('${safe(b.id)}',this.value)">

 ${["Pending","Approved","Rejected","Checked-
in","Completed","Cancelled"].map(s=>`<option
${b.status===s?"selected":""}>${s}</option>`).join("")}

 </select></td></tr>`).join("")||'<tr><td colspan="6">No bookings received.</td></tr>';

}

function addCar(){

 const
name=document.getElementById("aname").value.trim(),price=Number(document.getE
lementById("aprice").value),location=document.getElementById("alocation").value.trim
();

 if(!name||price<=0||!location)return toast("Enter vehicle name, valid price and
location.");


cars.push({id:Date.now(),name,type:document.getElementById("atype").value,price,loc
ation,

 seats:Math.max(1,Number(document.getElementById("aseats").value)||1),

 emoji:document.getElementById("aemoji").value.trim()||" 🚗 ",available:true});


save();renderCars();renderAdmin();document.getElementById("aname").value="";docu
ment.getElementById("aprice").value="";

 toast("Vehicle added successfully.");

}

function deleteCar(id){

 if(!confirm("Delete this vehicle from the listing?"))return;

 cars=cars.filter(c=>c.id!==id);save();renderCars();renderAdmin();toast("Vehicle
deleted.");

}

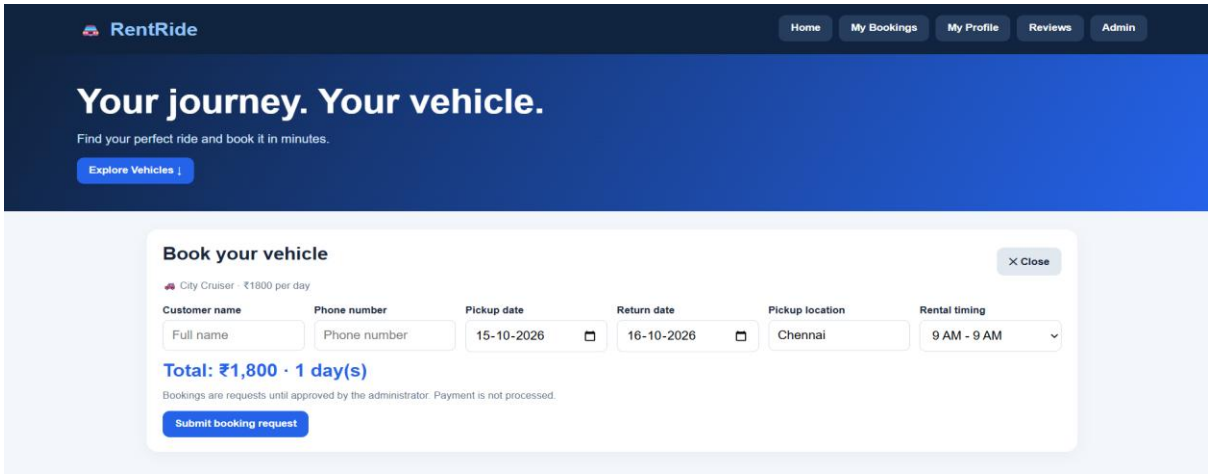
function updateStatus(id,status){

 const b=bookings.find(x=>x.id===id);if(!b)return;

 if(status==="Approved"){
```

```
const overlap=bookings.some(x=>x.id!==id&&x.carId===b.carId&&x.status==="Approved"&&b.start<x.end&&b.end>x.start);
if(overlap){renderAdmin();return toast("Cannot approve: another booking overlaps these dates.");}
}
b.status=status;save();renderAdmin();renderBookings();toast("Booking status updated.");
}
function exportBookings(){
const rows=[["ID","Customer","Phone","Vehicle","Start","End","Total","Status"],...bookings.map(b=>[b.id,b.name,b.phone,b.vehicle,b.start,b.end,b.total,b.status])];
const csv=rows.map(r=>r.map(v=>'"'+String(v).replace(/"/g,'""')+'"').join(",")).join("\r\n");
const url=URL.createObjectURL(new Blob([csv],{type:"text/csv;charset=utf-8"}));
const a=document.createElement("a");a.href=url;a.download="rentride-bookings.csv";a.click();URL.revokeObjectURL(url);
}
renderCars();renderReviews();
</script>
</body>
</html>
```

Output:



Result

Thus, the RentRide online vehicle rental system was developed successfully with vehicle searching, rental cost calculation, booking management, customer profiles, reviews, and an administrator dashboard.